

School of Artificial Intelligence and Big Data
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Prof. Stephen T. C. Wong and Prof. Xiaoyin Xu
Co-Editors-in-Chief
Computerized Medical Imaging and Graphics
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Dear Editors-in-Chief,

We submit the manuscript “**Pseudo-3D Heightfield Reconstruction from Monocular Endoscopic Video via Gradient-Anchored Geometric Constraints**” as a Research Article for consideration in *Computerized Medical Imaging and Graphics* (CMIG).

Why CMIG. The paper addresses a computational and visualization problem in endoscopic imaging: recovering a metric, renderable 3D surface from monocular (and, when available, stereo) surgical video. Collapsed or planar reconstructions cannot support augmented-reality overlay or image-guided measurement. The proposed 2.5-D heightfield is both a geometric representation and a visualizable mesh, which matches CMIG’s scope in reconstruction, visualization, endoscopy, and image-guided intervention.

What is new.

- A shared 2.5-D surface embedding: the same heightfield is registered for pose and queried for dense matching. Replacing that pose with a 2D essential matrix collapses the cloud, isolating the embedding rather than the fusion code.
- Shape is treated as a first-class criterion (λ_3/λ_1). Calibration patterns set the metric gauge and the mask; they are not the reconstructed primitive.
- An optional stereo variant that keeps the heightfield as a shape regularizer (AbsRel 0.0285 on the shared EndoNeRF 100-frame set).

Evidence. On an endoscopic chessboard benchmark, the proposed method and Stereo SGM are the only approaches that produce Normal 3D point clouds; the learned and SfM baselines collapse to cones or planes. Replacing heightfield-ICP poses with a 2D essential-matrix front-end collapses the cloud, isolating the heightfield signal. On SCARED, a 22-sequence stereo-laparoscopic set, and EndoNeRF, the method matches Stereo SGM in depth accuracy and supplies a confidence map for visualization. Four newly captured phantom sequences (different rig, calibrated PnP reference) show the same scale advantage on two sequences and a Normal 3D cloud on one; two sequences collapse, consistent with sparse metric anchors rather than a 32° rotation statistic inflated by planar-PnP flips.

Declarations. This manuscript is not under consideration elsewhere. All authors have approved the submission. The study uses public datasets and in-house bench-top phantom recordings; it does not involve new human-subject or animal experiments. The authors have no competing interests. Code, evaluation scripts, and result files are at <https://github.com/liranyang123456-commits/re3d-heightfield>. Full in-house sequences remain available from the corresponding author.

We thank you for considering this work.

Sincerely,

Ranyang Li, Junjun Pan, et al.